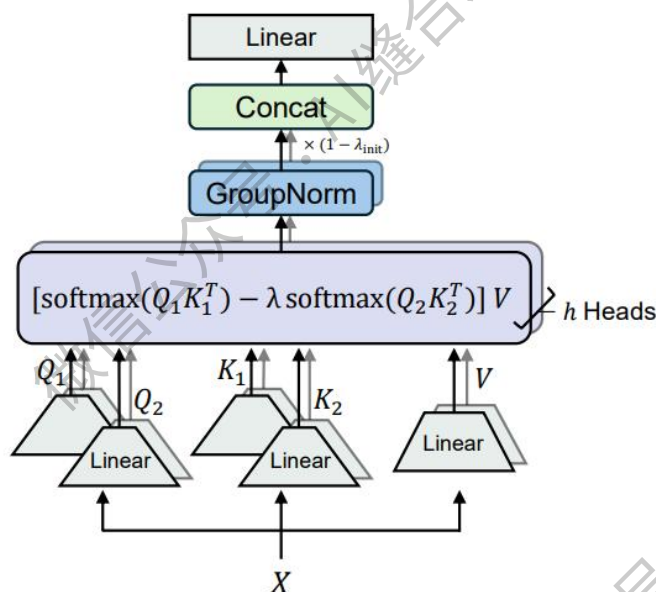


核心速览



```
def DiffAttn(X, W_q, W_k, W_v,  $\lambda$ ):  
    Q1, Q2 = split(X @ W_q)  
    K1, K2 = split(X @ W_k)  
    V = X @ W_v  
    # Qi, Ki: [b, n, d]; Vi: [b, n, 2d]  
    s = 1 / sqrt(d)  
    A1 = Q1 @ K1.transpose(-1, -2) * s  
    A2 = Q2 @ K2.transpose(-1, -2) * s  
    return  
        (softmax(A1) -  $\lambda$  softmax(A2)) @ V  
  
def MultiHead(X, W_q, W_k, W_v, W_o,  $\lambda$ ):  
    O = GroupNorm([DiffAttn(X, W_qi, W_ki,  
        W_v,  $\lambda$ ) for i in range(h)])  
    O = O * (1 -  $\lambda_{\text{init}}$ )  
    return Concat(O) @ W_o
```

论文题目: Differential Transformer

中文题目: 差分变压器

论文链接: <https://arxiv.org/pdf/2410.05258?>

代码链接: <https://github.com/microsoft/unilm/tree/master/Diff-Transformer>

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核心速览: 本文提出了一种名为 DIFF Transformer 的新型架构，通过引入差分注意力机制来减少无关上下文的干扰，从而提升大型语言模型在关键信息检索、幻觉缓解和上下文学习等任务中的表现。

论文内容详细解读:

<https://mp.weixin.qq.com/s/UoCh4CU6i-YvG3s9ZuOt2w>